

FloodGap

If Houston had funding for 20 new sensors, where are the 20 most ideal places to put them?

A sensor-siting model inspired by Savannah's CEAR Hub, applied to Houston

Starting Simple: Two Layers

NOAA tide data + social vulnerability

Layer 1: NOAA flood days

High-tide flooding days from NOAA tide stations.

Layer 2: Social vulnerability (CDC SVI)

Where flooding is equally likely, vulnerable communities suffer more and recover slower.

First lesson: NOAA stations are so sparse that all 174 coastal Georgia tracts shared ONE flood-day value. No variation = no ranking. The fix: add USGS elevation, which restores real tract-by-tract flood variation.

Then Inspired by the Paper

Tien et al. (2023), sensor siting in the same Savannah network, by a previous PhD student

Parameter	Paper (Savannah)	Houston adaptation	
1 • Compound flood	Single FEMA flood zone (tidal)	Tidal + stormwater + riverine combined	Core upgrade
2 • Vulnerability	Local DAPI index	CDC SVI at block-group level + note limits	Refined
3 • Infrastructure	Kernel density of facilities	Same, from OpenStreetMap	Reused
4 • Coverage	Expand existing 30 sensors	Spread 20 from scratch	Re-scoped
5 • Uncertainty	Reduce existing-network error	Simplified (no prior network)	Simplified

Problem 1: the Validation Isn't Valid

The existing sensors are 'where it was feasible', not 'where is ideal'

How the 48 sensors were really placed

- Installed 6 to 8 years ago, by feasibility: easy access, near a waterway, something to mount on
- Many strapped to fence posts; theft & stability were real concerns
- Deployment never fully completed (county infrastructure limits)

What my consistency check found

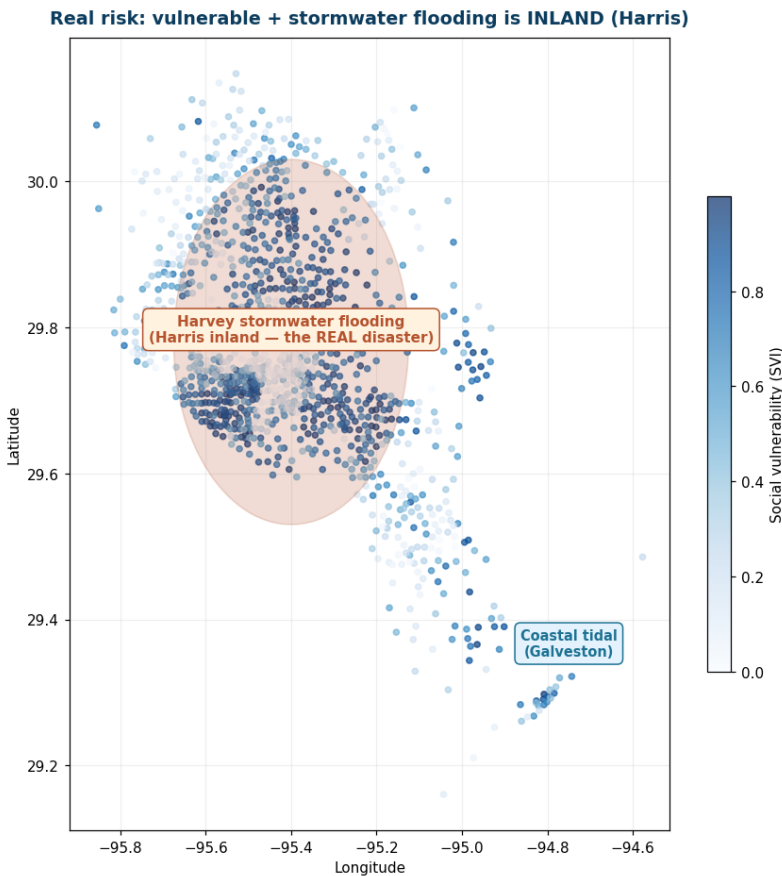
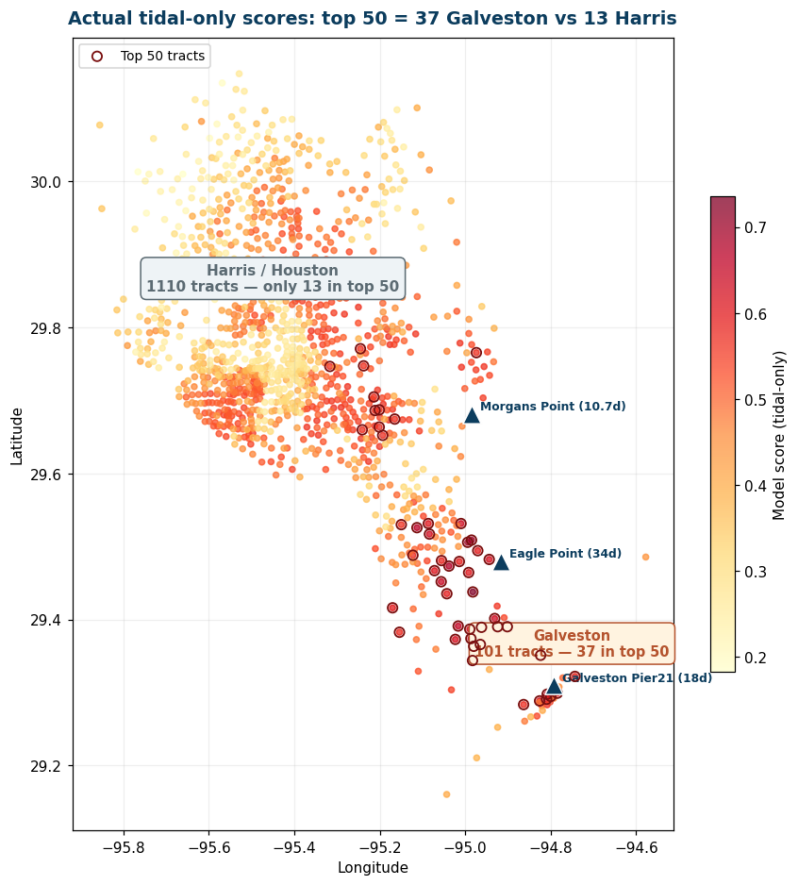
- Scored the 48 sensor sites with my model: they averaged only the 42nd percentile, below random
- Model favors flood + vulnerability; the sensors were driven almost purely by tidal feasibility
- **They can't serve as the 'correct answer' to fit weights against**

The reframe: the mismatch isn't a failed validation. It shows the real network is feasibility-limited and systematically misses high-risk, vulnerable places. That gap is what the model exposes, and it's the second meaning of 'FloodGap'.

Problem 2: Houston Doesn't Flood Like Savannah

Savannah is 90%+ tidal. Houston floods three ways.

Tidal-Only Model Misses Houston's Real Risk → Compound Flood Needed



Tidal-only result

30×

Galveston tracts entered the top 50
at 30× the rate of Harris County

*But Harvey's flooding was inland:
stormwater and bayous. The model was
blind to it.*

The Pivot

There is no 'correct' set of weights to validate

If you prioritize tidal risk → these are the top 20 places.

If you prioritize stormwater → those are the top 20.

If you prioritize vulnerable communities → different again.

Weights become policy options. The model shows decision-makers the trade-offs, and finds the places that matter under EVERY priority.

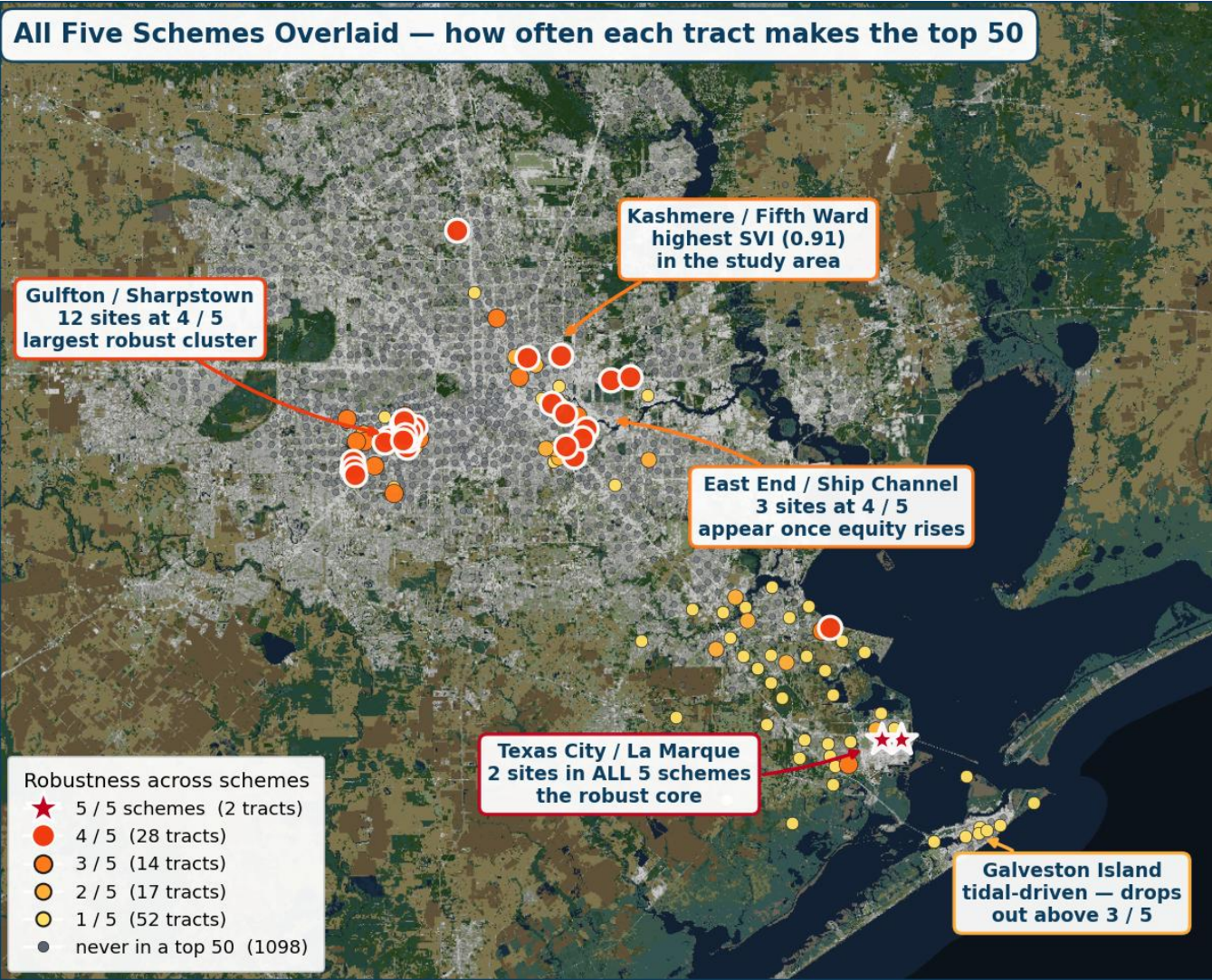
Five Weighting Schemes

Each scheme encodes a different priority, and the map changes with it

	Scheme	tidal/storm/river	flood/vuln	Harris/Galv	Leading regions
A	Tidal-only	100 / 0 / 0	60 / 40	3 / 47	Dickinson, Texas City, Galveston
B	Equal sub-weights	33 / 33 / 33	60 / 40	45 / 5	Gulfton, East End, Texas City
C	Harvey-informed	25 / 50 / 25	60 / 40	48 / 2	Gulfton, East End, Texas City
D	Flood-dominant	33 / 33 / 33	80 / 20	42 / 8	Gulfton, East End, Texas City
E	Equity-forward	33 / 33 / 33	50 / 50	46 / 4	Gulfton, Texas City, East End

Overlay All Five: the Robust Places Emerge

How many of the five schemes put each tract in the top 50



Robustness tally

5 / 5 schemes	2 tracts
4 / 5	28 tracts
3 / 5	14 tracts
2 / 5	17 tracts
1 / 5	52 tracts

Four robust clusters

Texas City / La Marque: in ALL 5 schemes

Gulfton / Sharpstown: 4 of 5, largest cluster

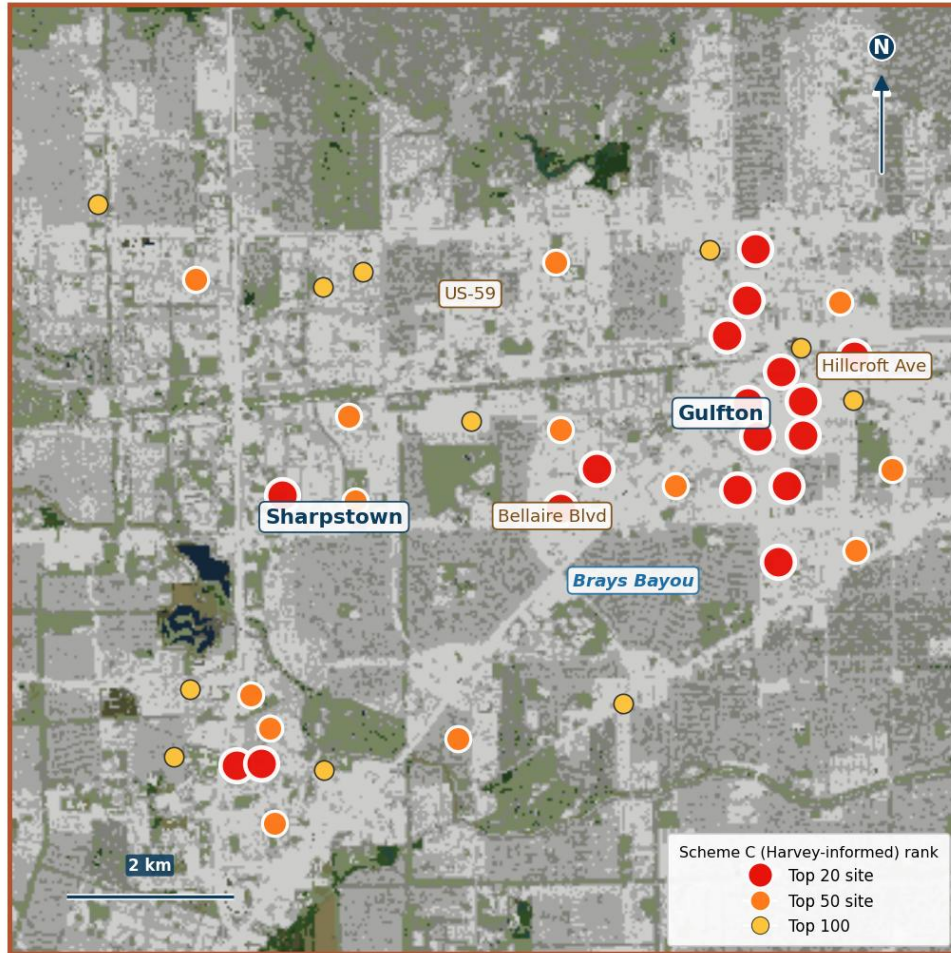
East End / Ship Channel: 4 of 5

Kashmere / Fifth Ward: highest SVI (0.91)

These are robust: high priority no matter how you weigh.

Zooming Into the Clusters

Each cluster has its own character, and its own sensor type



Gulfton under Harvey-informed weights: 17 of the top 20 sit here

Water-level sensor → tidal / riverine clusters

Texas City, East End, Dickinson. Mounted on bridges, culverts, dike structures over the water, like CEAR Hub's. Needs a waterway and clearance above base flood elevation.

Street-light sensor → stormwater clusters

Gulfton and Kashmere are inland, no waterway. Ultrasonic sensor points down from a light pole and measures street ponding. NYC FloodNet model, a few hundred dollars each.

Next Steps: Inside the Clusters

A concrete Houston research plan: from clusters to specific streets and poles

1

Fine-grain inside each Houston cluster

Re-run the scoring at block-group level inside Gulfton, Texas City, East End and Kashmere, then rank streets and intersections using 311 flooding complaints and HCFCF flood history.

2

Desktop pre-screen of candidate sites

For the top blocks, use Street View plus drainage records to shortlist 3 to 5 exact light poles or bridges per cluster, checking pole ownership (CenterPoint, TxDOT, HCFCF) and cellular signal.

3

Field reconnaissance in Houston

Visit the shortlisted sites with a structured checklist: street low points, stain lines, storm inlets, mounting structures. Site survey is my day job as a landscape architect.

4

Test against known gaps

Compare model picks with areas the team wanted to monitor but could not install in, and share results with Houston emergency management for real-world feedback.

The Bigger Picture

A transferable way to find the places that need protection most

This cluster-finding method isn't Houston-specific. Point it at Florida, or any coastal region, and it can surface the most at-risk counties and communities under any set of priorities.

For students, nonprofits, and agencies, that means quickly understanding a huge region: where to look first, where to protect first, where to measure first.

From one city's sensor question to a reusable screening tool. That is the bigger research direction.

FloodGap

The gap between where sensors are, and where they should be.



Thank you